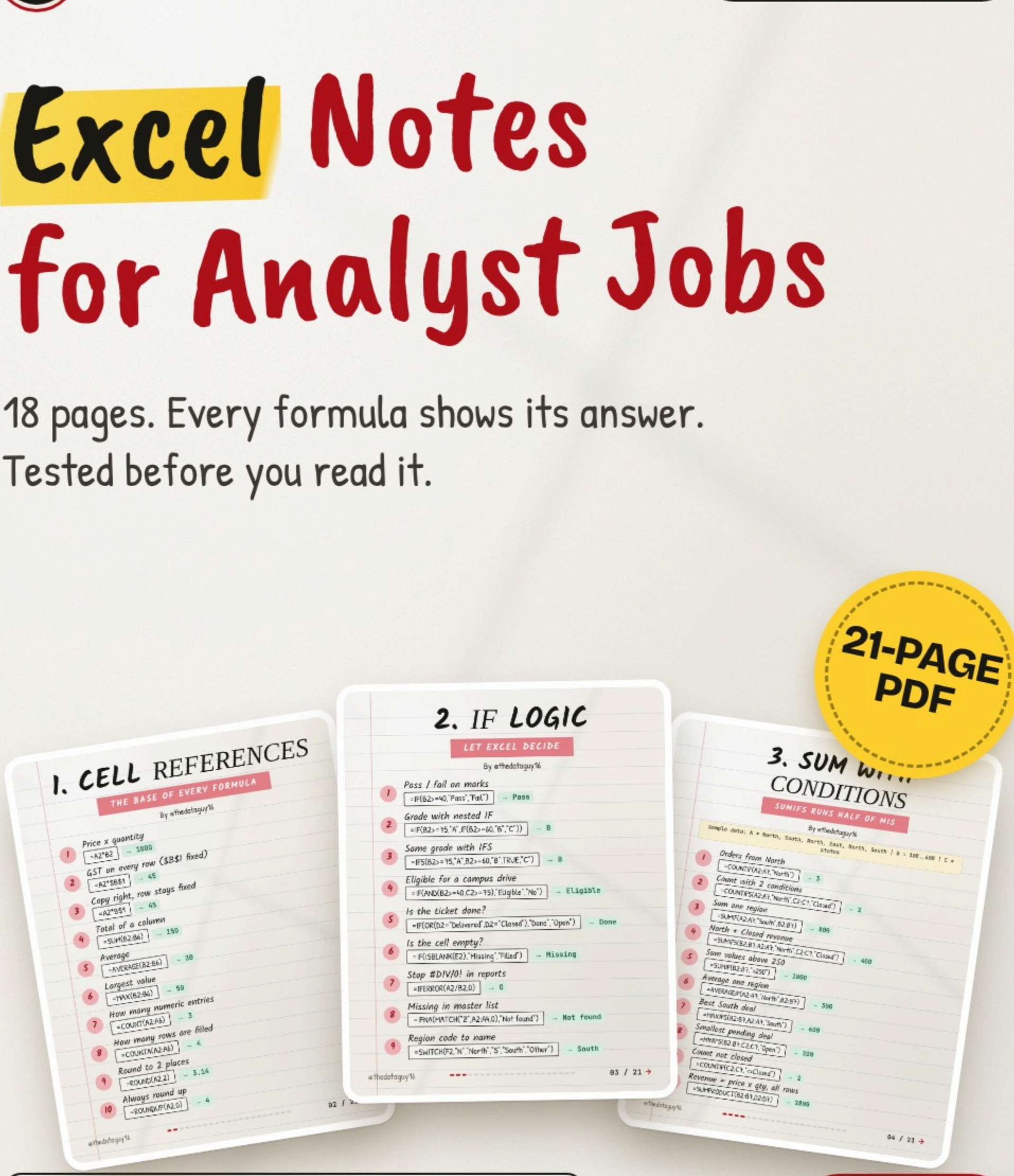
DG thedataguy16

SAVE FOR LATER

Excel Notes for Analyst Jobs

18 pages. Every formula shows its answer.
Tested before you read it.

21-PAGE PDF



Comment XLNOTES for the PDF

Swipe →

1. CELL REFERENCES

THE BASE OF EVERY FORMULA

By @thedataguy16

1. *Price x quantity*

```
=A2*B2
```
→ 1000

2. *GST on every row ($B$1 fixed)*

```
=A2*$B$1
```
→ 45

3. *Copy right, row stays fixed*

```
=A2*B$1
```
→ 45

4. *Total of a column*

```
=SUM(B2:B6)
```
→ 150

5. *Average*

```
=AVERAGE(B2:B6)
```
→ 30

6. *Largest value*

```
=MAX(B2:B6)
```
→ 50

7. *How many numeric entries*

```
=COUNT(A2:A6)
```
→ 3

8. *How many rows are filled*

```
=COUNTA(A2:A6)
```
→ 4

9. *Round to 2 places*

```
=ROUND(A2,2)
```
→ 3.14

10. *Always round up*

```
=ROUNDUP(A2,0)
```
→ 4

2. IF LOGIC

LET EXCEL DECIDE

By @thedataguy16

1 *Pass / fail on marks*

```
=IF(B2>=40,"Pass","Fail")
```
→ Pass

2 *Grade with nested IF*

```
=IF(B2>=75,"A",IF(B2>=60,"B","C"))
```
→ B

3 *Same grade with IFS*

```
=IFS(B2>=75,"A",B2>=60,"B",TRUE,"C")
```
→ B

4 *Eligible for a campus drive*

```
=IF(AND(B2>=40,C2>=75),"Eligible","No")
```
→ Eligible

5 *Is the ticket done?*

```
=IF(OR(D2="Delivered",D2="Closed"),"Done","Open")
```
→ Done

6 *Is the cell empty?*

```
=IF(ISBLANK(E2),"Missing","Filled")
```
→ Missing

7 *Stop #DIV/0! in reports*

```
=IFERROR(A2/B2,0)
```
→ 0

8 *Missing in master list*

```
=IFNA(MATCH("Z",A2:A4,0),"Not found")
```
→ Not found

9 *Region code to name*

```
=SWITCH(F2,"N","North","S","South","Other")
```
→ South

3. SUM WITH CONDITIONS

SUMIFS RUNS HALF OF MIS

By @thedataguy16

Sample data: A = North, South, North, East, North, South | B = 100..600 | C = status

1. *Orders from North*

```
=COUNTIF(A2:A7,"North")
```
→ 3

2. *Count with 2 conditions*

```
=COUNTIFS(A2:A7,"North",C2:C7,"Closed")
```
→ 2

3. *Sum one region*

```
=SUMIF(A2:A7,"South",B2:B7)
```
→ 800

4. *North + Closed revenue*

```
=SUMIFS(B2:B7,A2:A7,"North",C2:C7,"Closed")
```
→ 400

5. *Sum values above 250*

```
=SUMIF(B2:B7,">250")
```
→ 1800

6. *Average one region*

```
=AVERAGEIF(A2:A7,"North",B2:B7)
```
→ 300

7. *Best South deal*

```
=MAXIFS(B2:B7,A2:A7,"South")
```
→ 600

8. *Smallest pending deal*

```
=MINIFS(B2:B7,C2:C7,"Open")
```
→ 200

9. *Count not closed*

```
=COUNTIF(C2:C7,"<>Closed")
```
→ 2

10. *Revenue = price x qty, all rows*

```
=SUMPRODUCT(B2:B7,D2:D7)
```
→ 2800

4. LOOKUPS

VLOOKUP TO XLOOKUP

By @thedataguy16

Sample data: A2:A4 = 101, 102, 103 | B = Asha, Ravi, Neha | C = Pune, Indore, Jaipur | E2 = 102 | A5:B5 = 102, Ravi K | G2 = 72000, slab H2:I4 = 0 Low, 50000 Mid, 100000 High

1. *Name from employee ID*

```
=VLOOKUP(E2,A2:C4,2,FALSE)
```

→ Ravi

2. *City from employee ID*

```
=VLOOKUP(E2,A2:C4,3,0)
```

→ Indore

3. *Salary band from a slab table*

```
=VLOOKUP(G2,H2:I4,2,TRUE)
```

→ Mid

4. *Position in a list*

```
=MATCH(E2,A2:A4,0)
```

→ 2

5. *Value at a position*

```
=INDEX(B2:B4,2)
```

→ Ravi

6. *INDEX + MATCH, works anywhere*

```
=INDEX(B2:B4,MATCH(E2,A2:A4,0))
```

→ Ravi

7. *ID from a name (left lookup)*

```
=INDEX(A2:A4,MATCH("Neha",B2:B4,0))
```

→ 103

8. *XLOOKUP (Excel 365)*

```
=XLOOKUP(E2,A2:A4,B2:B4)
```

→ Ravi

9. *With a not-found value*

```
=XLOOKUP(999,A2:A4,B2:B4,"Not found")
```

→ Not found

10. *Last match, not first*

```
=XLOOKUP(E2,A2:A5,B2:B5,,0,-1)
```

→ Ravi K

5. CLEAN TEXT

MESSY NAMES, FIXED

By @thedataguy16

1. *Fix pasted names*
   `=TRIM(A2)` → ravi KUMAR
2. *Clean name for an offer letter*
   `=PROPER(TRIM(A2))` → Ravi Kumar
3. *Capital letters*
   `=UPPER(B2)` → PUNE
4. *Small letters*
   `=LOWER(B2)` → pune
5. *Invoice prefix*
   `=LEFT(C2,3)` → INV
6. *Invoice number*
   `=RIGHT(C2,4)` → 2041
7. *From the middle*
   `=MID(C2,5,2)` → 20
8. *Length*
   `=LEN(C2)` → 8
9. *Position of a dash*
   `=FIND("-",C2)` → 4
10. *Clean phone numbers*
    `=SUBSTITUTE(D2,"-","")` → 9820011111

6. JOIN AND SPLIT

NAMES, EMAILS, LABELS

By @thedataguy16

1 *Join with a space*

```
=A2&" "&B2
```
→ Asha Rao

2 *Same with CONCAT*

```
=CONCAT(A2," ",B2)
```
→ Asha Rao

3 *Join a list, skip blanks*

```
=TEXTJOIN(", ",TRUE,A2:A4)
```
→ Pune, Indore

4 *Username from email*

```
=LEFT(C2,FIND("@",C2)-1)
```
→ asha

5 *Company from email*

```
=MID(C2,FIND("@",C2)+1,50)
```
→ corp.in

6 *Number pasted as text*

```
=VALUE(E2)
```
→ 1250

7 *Percent label for a slide*

```
=TEXT(F2,"0.0%")
```
→ 25.6%

8 *Show a date your way*

```
=TEXT(G2,"dd-mmm-yyyy")
```
→ 05-Mar-2026

9 *Repeat a character*

```
=REPT("*",H2)
```
→ ****

7. DATES

REPORTS RUN ON DATES

By @thedataguy16

`Sample data: A2 = 15-Mar-2026 | B2 = 15-Oct-2026`

1. *Today's date*

```
=TODAY()>DATE(2026,1,1)
```
→ TRUE

2. *Year*

```
=YEAR(A2)
```
→ 2026

3. *Month number*

```
=MONTH(A2)
```
→ 3

4. *Month label for a pivot*

```
=TEXT(A2,"mmm")
```
→ Mar

5. *Month-end for MIS cut-off*

```
=EOMONTH(A2,0)
```
→ 31-Mar-2026

6. *Probation end date*

```
=EDATE(A2,3)
```
→ 15-Jun-2026

7. *Months of experience*

```
=DATEDIF(A2,B2,"m")
```
→ 7

8. *Working days to close a ticket*

```
=NETWORKDAYS(A2,B2)
```
→ 154

9. *Weekday (Mon = 1)*

```
=WEEKDAY(A2,2)
```
→ 7

10. *30 days later*

```
=A2+30
```
→ 14-Apr-2026

8. EXCEL 365 FORMULAS

NOT IN EXCEL 2016 OR 2019

By @thedataguy16

Sample data: A = Pune, Indore, Pune, Jaipur | B = Asha, Ravi, Neha, Imran | C = 300, 900, 700, 200

1 *Each city once*

`=UNIQUE(A2:A5)` → Pune, Indore, Jaipur

2 *Cities seen only once*

`=UNIQUE(A2:A5,,TRUE)` → Indore, Jaipur

3 *Count distinct cities*

`=ROWS(UNIQUE(A2:A5))` → 3

4 *Sort by sales, high first*

`=SORTBY(A2:C5,C2:C5,-1)` → Indore 900, Pune 700...

5 *Rows with sales over 500*

`=FILTER(A2:C5,C2:C5>500)` → Indore 900, Pune 700

6 *Same, with a fallback*

`=FILTER(A2:C5,C2:C5>5000,"None")` → None

7 *Two conditions (AND)*

`=FILTER(A2:C5,(A2:A5="Pune")*(C2:C5>500))` → Pune, Neha, 700

8 *Split at commas*

`=TEXTSPLIT(D2,",")` → Pune | Indore | Jaipur

9 *Split, skip empty bits*

`=TEXTSPLIT(D2,",",,TRUE)` → Pune | Indore | Jaipur

9. PERCENT AND GROWTH

EVERY MIS REVIEW ASKS THESE

By @thedataguy16

Sample data: B2:B5 = 100, 200, 300, 400 growth rows use B2 = last month, C2 = this month

1. *Region share of total*

```
=B2/SUM($B$2:$B$5)
```
→ 10%

2. *Growth vs last month*

```
=(C2-B2)/B2
```
→ 25%

3. *Running total*

```
=SUM($B$2:B4)
```
→ 600

4. *Rank the sales team*

```
=RANK.EQ(B3,$B$2:$B$5)
```
→ 3

5. *Second highest*

```
=LARGE(B2:B5,2)
```
→ 300

6. *Lowest*

```
=SMALL(B2:B5,1)
```
→ 100

7. *Average yearly growth (CAGR)*

```
=(C2/B2)^(1/3)-1
```
→ 10%

8. *Percent as a label*

```
=ROUND(B2/C2*100,1)&"%"
```
→ 33.3%

9. *Growth when last month was 0*

```
=IFERROR((C2-B2)/B2,"")
```
→ (blank)

10. SPOT DIRTY DATA

CHECK BEFORE YOU ANALYSE

By @thedataguy16

1. *Duplicate invoice?*

```
=COUNTIF($A$2:$A$6,A2)>1
```
→ TRUE

2. *Label duplicates*

```
=IF(COUNTIF($A$2:$A$6,A2)>1,"Duplicate","")
```
→ Duplicate

3. *Is it blank?*

```
=ISBLANK(B2)
```
→ TRUE

4. *Is it a real number?*

```
=ISNUMBER(C2)
```
→ FALSE

5. *Number with spaces around it*

```
=VALUE(TRIM(C2))
```
→ 125

6. *Remove line breaks*

```
=CLEAN(D2)
```
→ PuneWest

7. *Count blank cells*

```
=COUNTBLANK(B2:B6)
```
→ 2

8. *Extra spaces hiding*

```
=LEN(F2)-LEN(TRIM(F2))
```
→ 2

9. *Exact match incl. case*

```
=EXACT(G2,H2)
```
→ FALSE

11. Pivot tables, step by step

by @thedataguy16

#	Step	How
1	Turn data into a table	Click inside the data, Insert > Table
2	Insert the pivot	Insert > PivotTable > New Worksheet
3	Sales by region	Drag Region to Rows, Sales to Values
4	Split by month	Drag Month to Columns
5	Count instead of sum	Value Field Settings > Count
6	Show % of total	Show Values As > % of Grand Total
7	Group dates by month	Right-click a date > Group > Months
8	Filter one region	Drag Region to Filters
9	Clickable filter buttons	PivotTable Analyze > Insert Slicer
10	Biggest first	Right-click a value > Sort > Largest to Smallest
11	Show only the best 5	Row label filter > Value Filters, set it to 5
12	New data added	Right-click the pivot > Refresh
13	Blanks should show 0	PivotTable Options > For empty cells show: 0
14	Make a chart from it	PivotTable Analyze > PivotChart
15	See rows behind a number	Double-click that number
16	Flat, report-style layout	Design > Report Layout > Show in Tabular Form

12. CONDITIONAL FORMATTING

RULES THAT USE A FORMULA

By @thedataguy16

1. Sales above 1000

   `=$C2>1000` → TRUE

2. Pending orders

   `=$D2="Pending"` → TRUE

3. Duplicates in column A

   `=COUNTIF($A:$A,$A2)>1` → TRUE

4. Highest value

   `=C2=MAX($C$2:$C$10)` → TRUE

5. Every other row

   `=MOD(ROW(),2)=0` → TRUE

6. Overdue dates

   `=$E2<TODAY()` → TRUE

7. Empty cells

   `=ISBLANK($B2)` → TRUE

8. Between 500 and 999

   `=AND($C2>=500,$C2<1000)` → TRUE

9. Weekend dates

   `=WEEKDAY($E2,2)>5` → TRUE

10. Text contains delhi

    `=ISNUMBER(SEARCH("delhi",$F2))` → TRUE

13. MIS REPORT FORMULAS

MONTHLY REVIEW, SORTED

By @thedataguy16

Sample data: A = region | B = Jan, Jan, Feb, Mar | C = 100, 200, 300, 400

1. *January total*

```
=SUMIFS(C:C,B:B,"Jan")
```
→ 300

2. *North in January*

```
=SUMIFS(C:C,A:A,"North",B:B,"Jan")
```
→ 100

3. *Orders in a month*

```
=COUNTIFS(B:B,"Mar")
```
→ 1

4. *Share of one region*

```
=SUMIF(A:A,"North",C:C)/SUM(C:C)
```
→ 80%

5. *Target met?*

```
=IF(D2>=E2,"Met","Missed")
```
→ Met

6. *Target achievement*

```
=D2/E2
```
→ 120%

7. *Month on month*

```
=D3/D2-1
```
→ 20%

8. *Weighted average price*

```
=SUMPRODUCT(F2:F4,G2:G4)/SUM(G2:G4)
```
→ 22.5

9. *Year to date*

```
=SUM($D$2:D4)
```
→ 350

10. *Gap to target*

```
=MAX(0,E2-D2)
```
→ 20

14. Which chart, when

by @thedataguy16

#	You want to show	Use
1	Change over months	Line chart
2	Compare a few categories	Column chart
3	Long category names	Horizontal bar chart
4	Rank many items	Bar chart, sorted
5	Share of a whole, 2 to 4 parts	Pie or doughnut
6	Share, month by month	Stacked column
7	Two measures, different scales	Combo chart with a secondary axis
8	Link between two numbers	Scatter chart
9	How values are spread	Histogram
10	Ups and downs to a final total	Waterfall chart
11	One headline number	A big number with a label
12	Actual vs target	Column with a target line
13	Few causes, most of the effect	Pareto chart
14	Sales by state	Map chart (Excel 365)

15. Shortcuts that save time

by @thedataguy16

#	Do this	Keys (Windows)
1	Make a table	`Ctrl + T`
2	Filters on or off	`Ctrl + Shift + L`
3	AutoSum	`Alt + =`
4	Flash Fill	`Ctrl + E`
5	Quick chart	`Alt + F1`
6	Chart on a new sheet	`F11`
7	Format Cells	`Ctrl + 1`
8	Add $ to a reference	`F4`
9	Fill down	`Ctrl + D`
10	Edit the cell	`F2`
11	New line inside a cell	`Alt + Enter`
12	Select the column	`Ctrl + Space`
13	Select the row	`Shift + Space`
14	Insert today's date	`Ctrl + ;`
15	Paste Special	`Ctrl + Alt + V`
16	Jump to the edge of data	`Ctrl + Arrow key`

16. Errors and how to fix them

by @thedataguy16

#	Error	What it usually means
1	#N/A	Lookup value not found. Check spaces, text vs number.
2	#VALUE!	Text where a number was expected.
3	#REF!	The formula points to a deleted cell.
4	#DIV/0!	Dividing by zero or a blank cell.
5	#NAME?	Misspelt function, missing quotes, or not in your version.
6	#SPILL!	A 365 formula has no room. Clear the cells below.
7	#NUM!	An impossible number, like a bad date sum.
8	#####	Column too narrow. Widen it.
9	Total looks wrong	Numbers stored as text. They sit on the left.
10	VLOOKUP picks a wrong row	Last argument missing. Use FALSE.
11	Formula not updating	Formulas > Calculation Options > Automatic.
12	Circular reference	The formula uses its own cell.

17. Interview questions, short answers

by @thedataguy16

#	Question	Answer
1	VLOOKUP vs XLOOKUP?	XLOOKUP can look left, has a not-found value and is exact by default.
2	What does $ do in $A$1?	Locks the column and row when you copy.
3	COUNT vs COUNTA?	COUNT: numbers only. COUNTA: any filled cell.
4	Remove duplicates?	Data > Remove Duplicates, or UNIQUE in 365.
5	SUMIF vs SUMIFS?	SUMIF: one condition. SUMIFS: many, sum range first.
6	What is a pivot table?	A drag-and-drop summary. No formulas needed.
7	Find duplicates, keep data?	COUNTIF(range, cell)>1 or highlight rules.
8	What is Flash Fill?	Ctrl + E. Excel repeats the pattern you typed.
9	Why INDEX + MATCH?	Looks both ways and survives inserted columns.
10	How do you handle errors?	Fix the cause, then IFERROR or IFNA.
11	Relative vs absolute?	Relative moves when copied. Absolute stays.
12	What is data validation?	Rules for what a cell accepts, like a drop-down.
13	Freeze the header row?	View > Freeze Panes > Freeze Top Row.
14	What is Power Query for?	Repeatable cleaning: import, clean, combine, refresh.

18. Practise these 12 tasks

by @thedataguy16

#	Task	You will use
1	Total sales by region and month	`SUMIFS or a pivot`
2	Fill a customer's city from another sheet	`XLOOKUP or INDEX + MATCH`
3	Clean 500 messy names	`TRIM, PROPER`
4	Split emails into name and domain	`LEFT, MID, FIND`
5	Flag duplicate invoices	`COUNTIF`
6	Grade students into A/B/C	`IFS`
7	Month-on-month growth table	`Percent formulas`
8	Overdue payments in red	`Conditional formatting`
9	Best 5 products by revenue	`SORTBY or a sorted pivot`
10	Working days per order	`NETWORKDAYS`
11	One-page sales dashboard	`Pivot + slicer + chart`
12	Target vs actual report	`IF, percent, column chart`

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